PREDICTIVE CHURN ANALYSIS

Customer Churn Prediction Using Logistic Regression

Project Report
Dataset: customer_churn_model_v2.csv
Implementation: Python in Visual Studio Code

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Course / Internship: _____
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1. Abstract

Customer churn occurs when a customer stops using a company's product or service. Predicting churn can help an organization identify customers who may be at risk of leaving and plan suitable retention activities. This project builds a supervised machine-learning classification pipeline using Logistic Regression. The workflow includes dataset inspection, missing-value handling, categorical feature encoding, numerical feature scaling, an 80:20 train-test split, model evaluation, ROC-curve generation, and export of customer-level churn risk predictions. On the test set, the model achieved a precision of 0.5763, recall of 0.5075, F1-score of 0.5397, ROC-AUC of 0.6808, and accuracy of approximately 64.38%.

2. Introduction

Customer retention is important because losing customers can affect revenue and long-term business growth. Machine learning can use historical customer information to estimate the likelihood that a customer will churn. This project applies Logistic Regression to classify customers into churn and non-churn groups and produces a churn-risk probability for each customer in the test set.

3. Objectives

- Inspect and prepare the customer churn dataset.
- Encode categorical variables using One-Hot Encoding.
- Scale numerical features using MinMaxScaler.
- Split the data into 80% training data and 20% testing data.
- Train a Logistic Regression classification model.
- Evaluate the model using Precision, Recall, F1-score, ROC-AUC, a classification report, and a confusion matrix.
- Plot and save the ROC curve.
- Export predicted churn labels and churn-risk scores to a CSV file.

4. Dataset Description

The project uses the provided file customer_churn_model_v2.csv. The dataset contains customer-related fields and a target column named Churn. The target values are interpreted as Yes (customer churned) and No (customer did not churn). CustomerID is retained for identifying rows in the exported predictions but is excluded from the model inputs.

The dataset was reported as containing 800 rows and 13 columns before preprocessing. The model evaluation used 160 test records, consistent with an 80:20 split of 800 records after target values were checked.

5. Methodology

5.1 Data preprocessing

The dataset was loaded with pandas and inspected for column names and missing values. Rows with missing target labels were removed. Numerical columns were imputed using the median and scaled using MinMaxScaler. Categorical columns were imputed using the most frequent value and transformed using OneHotEncoder. The preprocessing steps

were placed inside a scikit-learn pipeline so they were fitted on the training data rather than the test data.

5.2 Train-test split

The dataset was divided into 80% training data and 20% testing data using train_test_split. A fixed random_state of 42 was used for reproducibility, and stratification was applied to preserve the target-class proportions across the two subsets.

5.3 Model selection and training

Logistic Regression was selected as the classification algorithm. It learns a relationship between customer features and the churn label and outputs class predictions as well as estimated probabilities. The model was trained using the training subset.

5.4 Evaluation metrics

- Precision: the proportion of customers predicted to churn who actually churned.
- Recall: the proportion of actual churners correctly identified by the model.
- F1-score: the harmonic mean of precision and recall.
- ROC-AUC: measures how well the model ranks churners above non-churners across classification thresholds.
- Accuracy: the proportion of all test predictions that were correct.

6. Results

The model produced the following evaluation results on the test set:

Metric	Score	Percentage / interpretation
Precision (churn = Yes)	0.5763	57.63%
Recall (churn = Yes)	0.5075	50.75%
F1-score (churn = Yes)	0.5397	53.97%
ROC-AUC	0.6808	Moderate discrimination
Accuracy	0.6438	Approximately 64.38%

6.1 Classification report

Class	Precision	Recall	F1-score	Support
No	0.67	0.73	0.70	93
Yes	0.58	0.51	0.54	67
Macro average	0.62	0.62	0.62	160
Weighted average	0.63	0.64	0.63	160

6.2 Confusion matrix

Rows represent actual classes and columns represent predicted classes.

Actual / Predicted	No	Yes
No	68	25
Yes	33	34

The model correctly predicted 68 non-churn customers and 34 churn customers. It incorrectly predicted churn for 25 customers who did not churn and missed 33 actual churners. The model therefore identified 34 of the 67 actual churners in the test set.

6.3 ROC curve

The ROC curve was generated and saved as outputs/roc_curve.png. The ROC-AUC score was 0.6808. This indicates that the model has some ability to distinguish churners from non-churners, but its performance is moderate and there is room for improvement. Insert the saved ROC curve image here if a printed or PDF report is required.

7. Prediction Export

The model exported test-set predictions to outputs/customer_churn_predictions.csv. The file includes CustomerID, Actual_Churn, Predicted_Churn, Churn_Risk_Score, and Churn_Risk_Percentage. The churn-risk score is the model-estimated probability of churn, not a guarantee that a customer will leave.

8. Discussion

The model achieved an ROC-AUC of 0.6808 and an F1-score of 0.5397 for the churn class. Its recall of 0.5075 means that it identified about half of the actual churners in the test set. The confusion matrix shows that 33 churners were incorrectly classified as non-churners. For a retention-focused use case, these missed churners are important because they could be overlooked by a retention team. The current model can serve as a baseline, but its results should be improved and validated before any real business use.

9. Limitations and Future Improvements

- Compare Logistic Regression with Decision Tree and Random Forest models.
- Tune model hyperparameters using cross-validation.
- Investigate class imbalance and test suitable class weights or resampling methods where appropriate.
- Choose a classification threshold based on the relative cost of missed churners and unnecessary retention offers.
- Check probability calibration before interpreting risk scores as reliable probabilities.
- Evaluate the model on additional or future data before using it for real customer-retention decisions.

10. Conclusion

This project implemented a complete predictive churn classification pipeline using Python and Logistic Regression. The workflow covered data preprocessing, categorical encoding, numerical scaling, an 80:20 train-test split, model training, evaluation, ROC-curve generation, and export of customer-level churn predictions. The model achieved a precision of 0.5763, recall of 0.5075, F1-score of 0.5397, ROC-AUC of 0.6808, and accuracy of approximately 64.38%. These results provide a baseline for churn prediction, while further model comparison and tuning are recommended to improve performance.

11. Project Files

- churn_analysis.py — Python source code.
- customer_churn_model_v2.csv — input dataset.
- outputs/customer_churn_predictions.csv — exported test-set predictions.
- outputs/roc_curve.png — ROC curve visualization.
- outputs/customer_churn_model.pkl — saved trained pipeline.

Note: The results in this report are based on the metrics visible in the project run. Confirm that the dataset size and output files match your final run before submission.